

A Triage-and-Audit Resource for the 9,408 Undeciphered Classes of HUST-OBC, with 279 Verified Metadata Corrections

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Abstract

Of the $\sim 4,500$ attested oracle bone script (OBS) characters, roughly 3,000 remain undeciphered, and recent machine-learning datasets—notably HUST-OBC—expose thousands of “undeciphered” character classes as a natural target space for computational decipherment. We release a **triage-and-audit resource** over all **9,408 undeciphered classes of HUST-OBC** that attaches to each class, from the authoritative Yinqi Wenyuan platform, (i) its most shape-similar official glyph code and that code’s official reading and status, and (ii) whether the authoritative compendium *Jiaguwenzi Gulin* (*Jitilin*) already contains a discussion entry for it. A construction-guaranteed identity mapping additionally yields **279 verified metadata corrections**—classes labelled “undeciphered” in HUST-OBC that in fact already carry an official reading—each re-verified live against the platform API (279/279). The resource makes explicit a property that is otherwise invisible: a majority of “undeciphered” classes are *already discussed* in the literature, so the resource functions as a filtered target space that separates genuinely-unclaimed characters from already-studied ones. All records are reproducible from public, login-free endpoints; no copyrighted scans or fonts are redistributed.

1 Background & Summary

The digitization of OBS has produced large labelled image datasets (HUST-OBC, HWOBC, OBC306) that invite computational decipherment. A recurring but under-examined assumption is that a dataset’s “undeciphered” label marks a character as an open research target. It does not: the label means only that no single consensus transcription exists, which is distinct from “not yet studied.” Many characters have been discussed—often with competing readings—for a century without consensus. A pipeline that treats the whole undeciphered set as targets therefore wastes effort re-deriving known arguments.

This resource operationalizes that distinction at dataset scale. For every one of the 9,408 HUST-OBC undeciphered classes we record two orthogonal official-code signals and a discussion-presence flag drawn from the Yinqi Wenyuan platform, which digitizes the *Jitilin* (a comprehensive exegetical compendium), the *Leizuan* concordance, and standard glyph dictionaries. The resulting table lets any user (a) obtain, per class, the most shape-similar known character and whether that character is itself deciphered, and (b) filter to classes with *no* existing *Jitilin* discussion—the subset most likely to be genuinely unclaimed. As a high-confidence by-product, a construction-guaranteed identity mapping over the Y+H classes surfaces 279 classes whose “undeciphered” label is stale, providing an immediately actionable dataset erratum. The resource’s reuse value is thus twofold: a **prioritized, discussion-aware target space** for decipherment research, and a **verifiable correction set** for dataset maintainers.

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Key figures. 9,408 classes (source groups: L 4,444, X 2,288, Y+H 2,676). Top-1 shape match is to a deciphered code for 4,409 classes and to an undeciphered code for 4,999. A *Jitilin* entry exists for the top-1 match of 6,229 classes (66.2% of all classes; 66.7% of resolved lookups), false for 3,107, and unresolved for 72 (API failure). 279 classes are verified metadata corrections (266 plain readings, 11 loan/variant, 2 tentative).

2 Methods

Source datasets. Glyph images and class labels are from HUST-OBC [1, CC BY 4.0]. Official readings, glyph codes, and exegetical cross-references are from the Yinqi Wenyuan platform [9], queried through its public, login-free endpoints (documented in the code repository).

Shape-retrieval signal (all 9,408 classes). A ResNet-18 encoder, fine-tuned on the 1,588 deciphered HUST-OBC classes (penultimate 512-d features), encodes each undeciphered class by mean-pooling its variant images; cosine similarity against a gallery of rendered official glyphs yields the top-1 official code and similarity per class. The encoder is a standard, replaceable component; its held-out retrieval accuracy (Technical Validation) establishes the signal’s quality but is not itself a contribution.

Constructive-identity signal (2,676 Y+H classes). HUST-OBC Y+H directory names equal the last five hexadecimal digits of official glyph codes (e.g. Y+H/60007 → U60007). This gives a by-construction (not similarity-inferred) mapping to the official code, its reading, and its status. X and L names are source-book serials and have no such mapping.

Discussion-presence flag. For each relevant official code we query `getzxbybm` and record whether the response links a *Jitilin* entry (`GLLX=1`). Requests are retried; unresolved codes after retries are marked `unknown-if-fetch-failed`.

Corrections set. Classes whose constructive-identity code carries a nonempty official reading while HUST-OBC labels them undeciphered constitute the 279 corrections; all were re-queried live (279/279 returning the expected reading).

3 Data Records

The resource is deposited on Zenodo (DOI: [10.5281/zenodo.21290640](https://doi.org/10.5281/zenodo.21290640)) and comprises:

1. **trriage_resource.csv / .json — 9,408 rows, one per undeciphered class.** Columns (full definitions in `trriage_resource_data_dictionary.md`): `hustobc_class`, `source_group`; the constructive-identity group `identity_official_code`, `identity_official_reading_jtz`, `identity_official_status` (populated for Y+H only); the shape-retrieval group `top1_official_code`, `top1_similarity`, `top1_official_reading_jtz`, `top1_official_status`, `top1_has_jitilin_entry`; and the flag `is_verified_correction`.
2. **yh279_corrections.csv / .json — 279 rows.** The verified corrections subset, with `id`, `class`, `official code`, `reading`, `reading_type` (plain / loan_or_variant / tentative), `evidence source`, `per-row reproduction URL`, `fetch date`, and `verifier`.

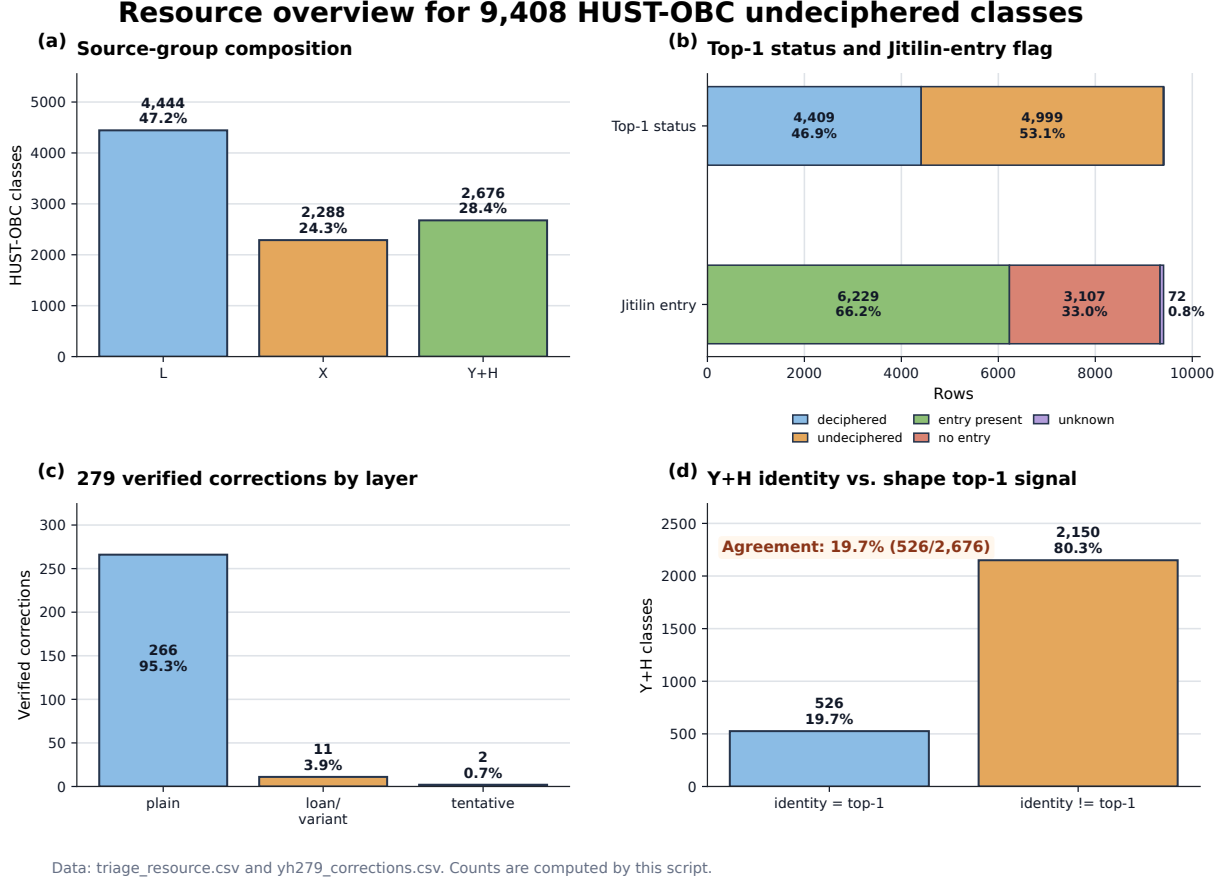


Figure 1: **Overview of the resource over 9,408 HUST-OBC undeciphered classes.** (a) Source-group composition (L 4,444; X 2,288; Y+H 2,676). (b) Top-1 shape-match status (deciphered 4,409 / undeciphered 4,999) and *Jitilin*-entry flag (entry present 6,229 = 66.2%; no entry 3,107; unresolved 72). (c) The 279 verified corrections by reading type (plain 266; loan/variant 11; tentative 2). (d) For the 2,676 Y+H classes, the constructive-identity code and the shape top-1 code agree only 19.7% of the time, quantifying the font-vs-handwriting domain gap. All counts are computed directly from the released tables.

3. **Provenance artifacts.** `trriage_resource_api_cache.json` (per-code API responses and retry outcomes), `trriage_resource_summary.json` (construction metadata and aggregate counts), and the deterministic builder script.

Two signals, not to be conflated. For Y+H rows the identity and shape signals are both present and should be read separately: the identity code is authoritative; the top-1 code is the independent shape signal. For X/L rows only the shape signal is available.

Table 1: Composition and per-class signals across the 9,408 undeciphered classes. Slice-level status columns are provided per row in the released table.

Slice	Classes	Top-1 deciph.	<i>Jitilin</i> present	Absent	Unresolved
L (Liu serials)	4,444	—	—	—	—
X (Xin serials)	2,288	—	—	—	—
Y+H (code-derived)	2,676	—	—	—	—
All classes	9,408	4,409 (46.9%)	6,229 (66.2%)	3,107 (33.0%)	72 (0.8%)

Table 2: The 279 verified metadata corrections, by reading type. All re-verified live (279/279).

Reading type	Count	Share	Meaning
Plain reading	266	95.3%	Official code carries a direct modern-character reading
Loan / variant	11	3.9%	Reading given with a 通假/异体 annotation
Tentative	2	0.7%	Reading marked provisional on the platform
Total	279	100%	All re-verified live (279/279)

4 Technical Validation

Corrections (279). Precision rests on two independent guarantees rather than a statistical estimate: (i) the identity mapping is exact by construction, and (ii) every entry was re-verified against the live API (279/279 returned the expected reading on 2026-07-06). The set was additionally submitted upstream to the HUST-OBC maintainers as an erratum (<https://github.com/Pengjie-W/HUST-OBC/issues/8>).

Shape-retrieval signal. Held-out retrieval accuracy validates the top-1 shape signal: on 4,219 query images never seen in training (drawn from 891 classes), Top-1 is 84.7% (Wilson 95% CI [83.6, 85.8]) and Top-10 95.5% ([94.9, 96.1]). A non-fine-tuned ImageNet ResNet-18 reaches only 26.2% / 50.4%, and a random baseline $\approx 0.05\%$, confirming the signal is driven by domain adaptation; across gallery sizes $\{4, 8, 16\}$ and two pooling schemes, Top-1 stays within 80–89% and Top-10 $\geq 94.8\%$.

Discussion-presence flag. Entry presence is reproducible and its aggregate is seed-robust: across five random samples of 200 undeciphered codes, the *Jitilin*-entry rate is a majority in every seed (55.5–71.5%, mean 63.2%), matching the whole-resource figure (66.7% of top-1 lookups).

Documented divergence (a property, not an error). For Y+H classes the identity code and the top-1 shape code agree only 19.7% of the time, quantifying the domain gap between rendered official fonts and handwritten/rubbing images. Users needing the authoritative code for Y+H should use the identity columns; the shape columns provide an independent view.

Known limitations. (i) Page-image “discussion-depth” proxies (ink density, column count) are exploratory and *not* validated against a ground-truth depth scale; all quantitative claims here rest on binary entry *presence*, not depth. Rigorous depth validation requires expert annotation and is left to future work. (ii) 72 classes have unresolved entry-presence (API failure), flagged as such. (iii) Post-*Jitilin* corpora (e.g. Huayuanzhuang 2003) trivially lack entries; users should stratify by attestation era before interpreting a false flag as “undiscussed.”

5 Usage Notes

- **Prioritized target space.** To find likely genuinely-unclaimed characters, filter `top1_has_jitilin_entry == false` (and, for Y+H, `identity_official_status == undeciphered`), optionally excluding post-*Jitilin*-only attestations. Conversely, `is_verified_correction == true` marks classes to relabel as deciphered.
- **Two-signal rule.** Never treat `top1_official_code` as the identity of a Y+H class; use `identity_official_code`. The two are deliberately separate.
- **Reproduction.** Every reading and flag is regenerable from the login-free endpoints documented in the repository; no copyrighted scans or fonts are included in the release.
- **Relationship to the methods paper.** The analysis motivating this resource (the pseudo-target problem, the discussion-depth triage framework, case studies) is reported in a companion methods manuscript, included as a preprint in the same data release (Zenodo DOI 10.5281/zenodo.21290640); this Descriptor documents the released data.
- **Minimal example (Python/pandas).** Load the resource and extract the highest-priority genuinely-unclaimed targets:

```
import pandas as pd
df = pd.read_csv("triage_resource.csv")
# Candidates with no existing Jitilin discussion (most likely truly unclaimed):
unclaimed = df[df.top1_has_jitilin_entry == "false"]
# Verified corrections to relabel as deciphered:
corrections = df[df.is_verified_correction == True]
# For Y+H classes, use the authoritative identity code, not the shape match:
yh = df[df.source_group == "Y+H"]
```

Data & Code Availability

Data package, builder, and companion preprint: Zenodo DOI [10.5281/zenodo.21290640](https://doi.org/10.5281/zenodo.21290640); release repository <https://github.com/shenzxc/hustobc-triage-resource>. Source materials remain © their publishers; only character codes, page identifiers, readings, and reproduction URLs are redistributed, under the platform’s public-access terms; HUST-OBC images are reused under CC BY 4.0.

Competing Interests

The author declares no competing interests. This work received no external funding (self-funded).

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